

TAN SIAO WAH

Artificial Intelligence graduate with research experience in Natural Language Processing and Computer Vision. Skilled in Python, PyTorch, TensorFlow, Hugging Face Transformers, model fine-tuning, data augmentation, and hyperparameter optimisation. Passionate about applying AI and machine learning to solve real-world problems and build intelligent systems.

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Portfolio: tansiaowah.github.io



WORK/PROJECT EXPERIENCE

Data Augmentation Enhanced Few-Shot Question Answering With Deep Learning

Master Research Project (2023-2025)

- Developed a dual-module data augmentation framework (QARR-QGF) to improve few-shot question answering performance in low-resource settings.
- Fine-tuned T5, BART, and Longformer transformer models on SQuAD, HotpotQA, and Natural Questions, achieving up to 82.3% F1 Score in the 16-shot setting.
- Designed data augmentation pipelines using rule-based question-answer replacement, paraphrase generation, and semantic filtering to improve training data diversity and model generalisation.
- Performed systematic hyperparameter tuning to optimise model performance.
- Developed an interactive Django-based QA web application integrating trained transformer models for context-aware question answering using custom text or Wikipedia articles.

Food Detection And Recognition Using Machine Learning Final Year Project (2021-2022)

- Conducted a comparative study of six object detection models (YOLOv5, Faster R-CNN, RetinaNet, and SSD) across three public datasets (45K+ images) for multi-class food detection.
- Developed end-to-end training pipelines, including dataset preprocessing, annotation conversion, image augmentation, hyperparameter tuning, and mean Average Precision (mAP) based evaluation, with the top-performing model achieved 0.931 mAP.

Artificial Intelligence Engineer Intern SmartPeep Sdn Bhd, Johor (Mar 2021-Jun 2021)

- Contributed to the Homefall system by integrating an object detection model with an environmental mapping approach in Golang, reducing false detections in human fall detection.
- Developed and trained human detection models with TensorFlow, including data collection, data preprocessing, data augmentation, model training and evaluation.
- Improved detection reliability by analysing failure cases and refining the model.

SKILLS

- Programming & Tools: Python, SQL (Postgres via Supabase, SQLAlchemy), Git, FastAPI, TypeScript, C++, HTML, CSS, Docker, n8n.
- Artificial Intelligence (AI) & Machine Learning (ML): PyTorch, Hugging Face, Natural Language Processing (NLP), Computer Vision (CV), TensorFlow, OpenCV, Few-Shot Learning, Object Detection, Data Preprocessing, Ground Truth Annotation, Data Augmentation, Model Fine-Tuning, Hyperparameter Optimisation, Model Evaluation.
- Languages: Mandarin, English, Bahasa Malaysia (speaking, reading and writing).

EDUCATION

Master of Science (Information Technology) (By Research)

Multimedia University, Melaka (2023-2025)

Bachelor of Information Technology (Hons.) Artificial Intelligence

Multimedia University, Melaka (2019-2022)

- CGPA: 3.94 (First Class Honours)

PUBLICATIONS

- QARR-QGF: A Dual-Module Data Augmentation Framework for Enhanced Few-Shot Question Answering
- QARR-FSQA: Question-Answer Replacement and Removal Pretraining Framework for Few-Shot Question Answering
- Food Detection and Recognition with Deep Learning: A Comparative Study